Abstract

Introduction:
International trade dynamics are influenced by different complex temporal and structural economic factors, making export growth prediction a non-trivial forecasting problem. This study formulates next-year export growth prediction as a supervised learning task using longitudinal country-level trade data of 34 years (1988–2021).

Dataset Description:
The analysis is done by using a publicly available dataset, it include 34 years (1988–2021) of country-level export and import statistics. The dataset includes several factors like trade values, balance indicators, and temporal features for multiple countries worldwide, enabling longitudinal economic modelling.

Methods:
The problem is formulated as a binary classification task, where I am focusing on export growth in the next year, predicted using Logistic Regression and Random Forest models. A time-based train-test split was applied to the dataset to avoid data leakage, and model performance was evaluated using accuracy, F1-score, and ROC-AUC.

Results:
The Random Forest model outperformed Logistic Regression for the dataset, demonstrating stronger predictive capability and better class separation in ROC analysis. Feature importance analysis revealed that trade balance and historical export trends are key drivers of export growth.

Discussion:
The findings from the dataset suggest that export dynamics exhibit non-linear patterns that are effectively captured by ensemble methods. While the dataset provides meaningful predictive power, limitations include missing macroeconomic variables and annual data granularity.

Conclusion:
Studying this dataset demonstrates that machine learning can effectively model export growth using long-term trade data. The proposed approach has practical applications in economic forecasting, trade policy analysis, and investment risk assessment.